

All the world's a stage, and all the agents merely players: Persona Driven Survey Response Generation

Arya Shah, Swaraj Bhanja, Suryansh Srivastava
Department of Computer Science and Information Management
Asian Institute of Technology
Pathumthani, Thailand
st125462, st125052, st14997@ait.ac.th

Abstract

One of the crucial frameworks for understanding various human behavior, attitude, and opinion is the conduction of surveys, which can contain responses across a number of diverse disciplines such as human-computer interaction, social science, market research and psychological traits, to name a few. However, the collection of diverse survey responses on a large scale remains to be a challenge across many domains which include difficulties in recruiting survey population, biases in response, and many other ethical considerations regarding privacy of the participants. Through this research, we introduce PersonaVerse, a novel framework that leverages LLMs (Large Language Models) which facilitate in the generation of synthetic yet realistic survey responses by encompassing the diverse world population which possesses multiple persona. Our approach creates LLM-powered agents who embody diverse and distinct behavioral, psychographic, and demographic characteristics derived from large-scale persona descriptions. These agents can then be leveraged into generating responses reflecting their unique persona attributes for various survey question modalities such as Likert scales, open-ended questionnaires, and multiple-choice questions. We incorporate a multifaceted evaluation framework in order to assess the diversity, quality, and consistency in the persona of the generated responses as compared to certain human generated benchmarks. We believe that our initial results shall demonstrate the novelty of PersonaVerse in producing diverse, contextually appropriate responses maintaining consistency with the assigned personas. We believe that our research contributes significantly to the growing field of synthetic data generation which possess high value of utility for researchers to not only augment human-collected survey data but also explore hypothetical demographic distributions, and simulate surveys before floating it to the human masses.

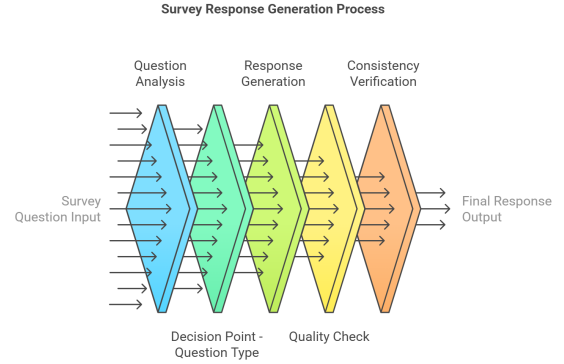


Figure 1: A high level illustration of the process of survey response generation facilitated by persona-based Large Language Models

1 Introduction

In order to understand human behavior, attitude, and preferences, the collection of survey responses for research purposes is fundamental. However, traditional survey methods possess limitations of being time-consuming, suffering from demographic biases inherent from the recruitment of survey population, being expensive, and may encounter ethical challenges related to sensitivity of topics and privacy of the survey responder. Moreover, researchers also need to explore hypothetical scenarios, and frequently test survey instruments before floating it to the masses- both being difficult to achieve with real human respondents.

Current approaches in this area include the collection of survey response and analysis through simple rule-based response generation, statistical modeling of existing data, and recently, leveraging large language models to generate responses in the absence of any persona grounding. These approaches often lack the diversity of authentic human responses which possess consistency in demographic, behavioral, and psychographic charac-

teristics across multiple questions.

We introduce the PersonaVerse framework, which facilitates the creation of persona-based agents capable of generating survey responses reflecting consistency in demographic and psychographic characteristics by leveraging large language models. When we approach grounding well-defined personas in response generation, it aims to create realistic and diverse synthetic responses compared to previous methods.

Our research offers several contributions, mainly: (1) method for instantiating LLM-Based agents for generation of responses to different question modalities possessing consistent persona characteristics; (2) a multifaceted framework for evaluating responses by assessment of quality, diversity and faithfulness in persona; and (3) an open source implementation for researchers to adapt according to various survey contexts. Although PersonaVerse does not replace human responses, it offers a complementary tool with the potential to improve survey methodologies for research and reduce the barriers to the conduct of large-scale, diversified survey studies. Throughout our research, our aim has been to answer the following formulated research questions, to answer, which we believe are crucial for the outcome of our study:

RQ 1: To what extent can LLM agents maintain consistency in persona characteristics while generating survey responses across a variety of question modalities and topics?

RQ 2: How effectively can a swarm of persona-based LLM agents generate survey responses as compared to human respondents in terms of diversity and representative distribution?

RQ 3: What practical applications and limitations pose in the synthetic responses generated by persona-infused LLM agents within research and commercial context?

2 Related Work

Synthetic Data Generation in itself has not been a relatively new topic or domain of research. Multiple studies have proposed a variety of methods for the creation of synthetic data. In this section, we shall be going through selected literature that explores this area closely and resonates with our topic of research.

2.1 Synthetic Data Generation

With the growing need to address challenges related to data scarcity, privacy concerns and need for diverse domain based robust AI training, synthetic data generation has proven to be a pivotal solution. There have been numerous innovations, early on, with survey methodologies such as Address-based Sampling (ABS), and mixed-mode data collection which claim to have improved representativeness with minimal bias but gaps still exist in fully addressing the issue of selection bias (Couper, 2017). Machine Learning applications have also leveraged generative models like GANs for synthetic data generation for the applications of classification tasks in the domain of healthcare and finance. While generative models perform this job effectively, challenges exist in capturing the complexity possessed by the real world (Abufadda and Mansour, 2021). Advances in the field of generative AI, especially with the advent of Large Language Models like GPT-3, the capability of enabling domain-specific synthetic data generation has become possible using techniques like prompt engineering and parameter efficient tuning (PEFT), to name a few. But the biggest concern is that of hallucination and ethical considerations (Guo and Chen, 2024). Making use of S-BERT-based synthetic data generation pipelines, semantic similarity between survey items has been shown to influence response patterns with results showing successful replication of human factors, but the limitation of handling negations as well as scaling across diverse survey types and modalities still prevails (Lehtonen et al., 2025)

2.2 LLMs for Text generation

With the rise of LLMs and their role in advancing Natural Language Processing tasks which includes synthetic data generation, a tool such as DataDreamer (Patel et al., 2024) provides a unified framework to address challenges of scalability and reproducibility in LLM-driven synthetic data generation, fine-tuning, and other similar research workflows. SynthIE (Josifoski et al., 2023) is another such innovative approach that exploits task asymmetry for the generation of high quality synthetic datasets. A similar framework, SynDG (Bao et al., 2023) generates synthetic grounded dialogues which are modeled through dialogue flows and filter strategies applied to enhance coherence and quality of the system. In the domain of text

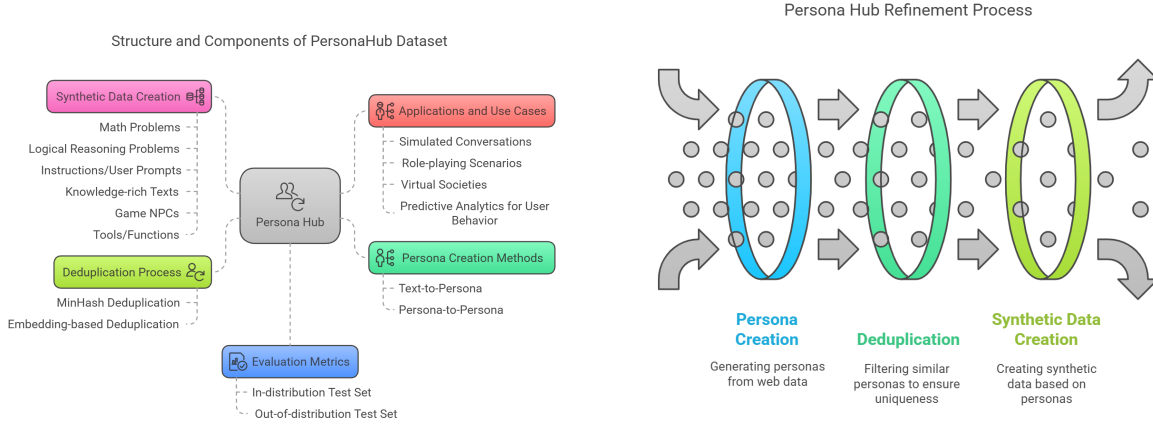


Figure 2: PersonaHub Dataset Structure, Components and Refinement Process

classification, the potential of LLMs has depicted that low-subjectivity tasks benefit more from synthetic data generation methods (Li et al., 2023). Authors of CodeLM use instruction tuning to tailor synthetic data for specific tasks using metadata encoding and refining the output iteratively, leading to superior results with diverse instructions. Finally, (Long et al., 2024) provide a comprehensive survey emphasizing the need for unified methodologies in the curation, generation and evaluation of such synthetic data for multiple NLP workflows. Some of the common gaps identified, include biases in LLM outputs, issues of coherence in dialogue and challenges in adapting to domain specific applications

2.3 Persona-Based Generation

The importance of modeling human-like behaviors and belief were explored soon after the prior research successfully made advancement in persona-driven survey response generation. A Generator-Critic architecture was introduced (Jandaghi et al., 2024), which efficiently facilitated creation of high-quality persona datasets addressing limitations in prior datasets such as Persona-Chat (Zhang et al., 2018). Although their approach demonstrated improvement in persona diversity, yet it struggled with intangible human traits like humor. In (Joshi et al., 2024), the authors cluster agents into personas which enhances truthfulness in noisy datasets, although challenges remained in defining the latent features that captured truthful personas reliably. Similarly, (Giorgi et al., 2024) evaluate explicit and implicit personas in order to replicate human biases. While explicit personas gave mixed

results in tasks such as belief generation, implicit personas completely failed to align with human level of subjectivity. The inability to consistently replicate biases in human responses across surveys designs was highlighted (Tjuatja et al., 2024), with RLHF-trained models introducing unwanted sensitivity despite marginal improvements in replicating human-like behavior. Finally, belief networks was proposed (Chuang et al., 2024) for the survey data to be enhanced in alignment with human opinions beyond role-playing on a demographic level. Although performance improved in terms of reduced errors, yet broader applicability failed to achieve due to its simplistic structure with limited scope.

3 Dataset

We build our dataset on top of the PersonaHub Dataset provided by Tencent AI Lab (Ge et al., 2024). In this section we discuss more about the dataset and how it helps us achieve our use case. The PersonaHub dataset introduces a person-driven approach for creation of synthetic data at scale making use of 1 billion personas curated from web data. Although the complete dataset has more than 1 billion personas, we shall be exploring only the tiny subset of 200,000 personas furthermore narrowing down to a couple of 1000s keeping in mind the computational resources we possess. As per the authors, these personas act as distributed carriers of the world’s information, enabling LLMs synthesize this dataset across various domains including math, logical reasoning, game NPC persona development, instructions, tools and knowledge-rich texts. For our research we utilize the detailed persona descriptions that include demographic in-

Method	Benefits	Gaps
Traditional Synthetic Data Generation	- Addresses data scarcity and privacy concerns effectively. - Enables domain-specific data creation using techniques like prompt engineering and PEFT. - S-BERT pipelines replicate human factors in surveys.	- Struggles with selection bias and accurately capturing real-world complexity. - Hallucination and ethical concerns remain significant challenges. - Limited ability to handle negations and scale across diverse survey types and modalities.
LLMs for Text Generation	- Tools like DataDreamer and SynthIE enhance scalability, reproducibility, and coherence. - Instruction tuning (e.g., CodeCLM) improves task-specific synthetic data generation. - Unified frameworks improve curation, generation, and evaluation of synthetic data.	- Biases in outputs, coherence issues in dialogues, and domain adaptation challenges persist. - Low subjectivity tasks benefit more, limiting broader applicability. - Ethical concerns and lack of consistency in domain-specific applications remain barriers.
Persona-Based Generation	- Generator-Critic architecture improves persona dataset quality and diversity. - Enhances truthfulness in noisy datasets through persona clustering. - Belief networks align survey data with human opinions beyond role-playing demographics.	- Struggles with intangible traits like humor and human subjectivity. - Challenges in defining latent features for reliable truthful personas persist. - Simplistic structure limits broader applicability across diverse scenarios.

Table 1: Qualitative comparison of methods discussed in Section 2.

formation, personality traits, interests, values, behavioral patterns, etc derived from this dataset. A structured format for these personas is created such that enables their use in effective conditioning of language model outputs. Rather than using the PersonaHub dataset out-of-the-box, we propose the personaVerse dataset, which follows a structured methodology for creating persona descriptions by giving them structure and then use it to adapt to different contexts and requirements. The Figure 2. shows the various components of the PersonaHub Dataset. Our PersonaVerse dataset is compiled with a diverse collection of survey questions spanning multiple domains such as:

- Consumer behavior
- Consumer preferences
- Social, Political attitudes
- Technology adoption and usage
- Healthcare

As per the modalities of the survey questions, we cover the following four most frequently used

modalities, providing the capability to generate responses for:

- Multiple Choice-type questions
- Likert scale based
- Ranking based
- Open-ended questions

3.1 PersonaVerse

The foundation of our research is the Personaverse dataset—a collection of 3,568 simulated survey responses across 10 domains, featuring 433 unique personas responding to 82 distinct questions derived from established benchmark surveys. This dataset was specifically constructed to enable the instruction-tuning of language models for survey response generation with the following characteristics:

1. Domain Diversity: Coverage spans Demographics, Healthcare, Education, Work Experience, Technology, Consumer Preferences, Finance, Social Issues, Environment, and Lifestyle

2. **Persona Variation:** Includes diverse demographic backgrounds, professions, interests, and perspectives to capture a wide range of response patterns
3. **Question Types:** Incorporates Likert-scale, open-ended, yes/no, and statement/agreement questions from established survey instruments
4. **Response Variation:** Contains responses of varying length, complexity, and sentiment to reflect authentic human response patterns

The dataset is organized in JSON format with each example following a consistent structure to facilitate efficient processing and model training. An examination of the dataset structure reveals careful attention to providing comprehensive conversational context for effective instruction tuning. Each example in the dataset contains a unique identifier, set of messages in ChatML format (system, user, assistant) and metadata including persona information, domain classification, and question details.

3.1.1 Domain Distribution and Characteristics

The dataset encompasses 10 distinct domains, selected to cover a broad spectrum of common survey contexts. Analysis of the domain distribution revealed a deliberate design choice to prioritize certain high-impact domains while maintaining reasonable coverage across all included categories:

1. **Demographics** (520 examples, 14.6%): Contains questions about age, gender, ethnicity, and other demographic factors
2. **Healthcare** (416 examples, 11.7%): Includes questions about health status, experiences with healthcare systems, and medical preferences
3. **Education** (416 examples, 11.7%): Focuses on educational background, learning experiences, and academic perspectives
4. **Work Experience** (400 examples, 11.2%): Covers employment history, workplace preferences, and career development
5. **Technology** (384 examples, 10.8%): Addresses technology adoption, digital preferences, and technical experiences
6. **Consumer Preferences** (368 examples, 10.3%): Examines purchasing behaviors, product preferences, and consumer attitudes
7. **Finance** (368 examples, 10.3%): Explores financial behaviors, attitudes toward money, and economic perspectives
8. **Social Issues** (264 examples, 7.4%): Covers perspectives on social policies, cultural attitudes, and community concerns
9. **Environment** (216 examples, 6.1%): Addresses environmental concerns, sustainability practices, and ecological perspectives
10. **Lifestyle** (216 examples, 6.1%): Examines personal habits, recreational activities, and quality of life factors

3.1.2 Persona Analysis and Diversity

The dataset features 433 unique personas designed to represent diverse demographic backgrounds, professions, interests, and perspectives. Our analysis of persona characteristics revealed a strong emphasis on reusing of persona- Most personas (71.4%) appear in only one domain, suggesting domain-specific persona design, while a smaller percentage (28.6%) appear across multiple domains, providing cross-domain consistency. The most common persona types include professionals (17.8%), educators (14.2%), students (12.5%), healthcare workers (11.3%), and technical specialists (10.7%). Clustering analysis applied to persona descriptions identified 8 distinct persona categories, each characterized by specific keyword patterns and demographic indicators. These clusters enable systematic analysis of response patterns across different demographic segments.

3.1.3 Survey Question Analysis

The dataset contains 82 unique questions distributed across the 10 domains. Analysis of these questions revealed important characteristics relevant to instruction tuning: The distribution includes open-ended questions (42.7%), Likert-scale questions (31.7%), yes/no questions (18.3%), and statement/agreement prompts (7.3%). Each domain exhibits distinct question patterns reflecting standard survey methodology in that field: healthcare questions emphasize patient experience and satisfaction metrics, technology questions focus on usage patterns and adoption factors, finance questions address financial literacy and behavioral economics.

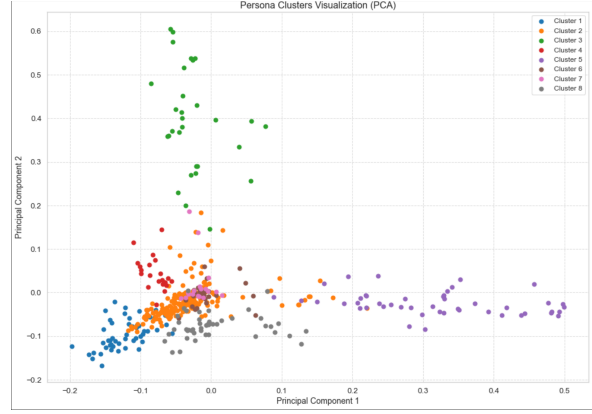
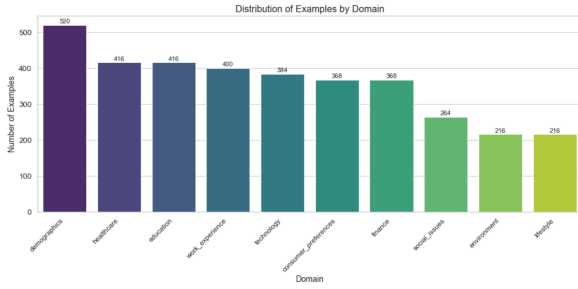


Figure 3: PersonaVerse Distribution of examples per domain(left) and the Persona Clusters(right)

Questions were derived from established benchmark surveys including HCAHPS (healthcare)(for Medicare & Medicaid Services , CMS), NSSE (education) (of Education Center for Postsecondary Research, 2024), TAM (technology adoption) (Davis, 1989), NEP Scale (environmental) (Dunlap et al., 2000), Gallup Q12 (workplace)(Buckingham and Coffman, 1999), WHOQOL-BREF (quality of life)(World Health Organization, 1998), ACSI (consumer satisfaction)(Fornell et al., 1996), World Values Survey (social attitudes)(World Values Survey Association, 2022), and FINRA (financial literacy)(Foundation, 2021).

4 Methodology

Our experimental approach focuses on instruction-tuning small language models (SLMs) for the specialized task of survey response generation using parameter-efficient techniques. We designed a comprehensive methodology that balances computational efficiency with model performance, enabling effective fine-tuning even with limited hardware resources.

4.1 Base Model Selection

For our primary experiments, we selected TinyLlama-1.1B-Chat as the base model based on several considerations:

1. **Parameter Efficiency:** At 1.1 billion parameters, TinyLlama offers a favorable balance between performance and computational requirements
2. **Pre-trained Capabilities:** TinyLlama includes chat-tuned capabilities that provide a strong foundation for response generation

3. **Instruction Following:** The model demonstrates basic instruction-following capabilities that can be enhanced through fine-tuning
4. **Memory Footprint:** The model’s relatively small size enables efficient training and inference on consumer-grade hardware
5. **For comparative analysis,** we also conducted experiments with Microsoft’s Phi-2 and Mistral-7B, Qwen-2 2B to evaluate performance across different model architectures and parameter scales. This multi-model approach provides insights into the scalability of our methodology. To ensure our approach is accessible to researchers with varying computational resources, we implemented a resource-adaptive configuration system that automatically adjusts training parameters based on available hardware. To make our approach accessible to researchers with varying computational resources, we developed a resource-adaptive configuration system that dynamically adjusts training parameters based on the available hardware. The system is implemented through a Config class, which begins with basic settings such as a fixed random seed, a base model (TinyLlama/TinyLlama-1.1B-Chat-v1.0), and an experiment name derived from the model. It then evaluates the computational environment to determine optimal configurations. If a GPU is available, the system checks its memory capacity: for high-end GPUs like the A100 with more than 30 GB of memory, it sets a batch size of 8, uses 8-bit precision, and avoids 4-bit precision; for mid-range GPUs like the RTX 3090 with over 12 GB of

Metric	Focus Area
General NLG Metrics	BLEU, ROUGE for assessing generation quality
Semantic Similarity Metrics	BERTScore and embedding-based similarity metrics to evaluate semantic alignment
Domain-Specific Metrics	Custom metrics for assessing domain-appropriate response characteristics
Response Diversity Metrics	Distinct-n metrics and lexical diversity measures to evaluate response variation
Persona Consistency	Metrics evaluating alignment between persona characteristics and response content

Table 2: Summary Table for adopted Evaluation Metrics

memory, it reduces the batch size to 4, uses 4-bit precision, and disables 8-bit precision; for lower-end GPUs like the RTX 2080Ti, it further reduces the batch size to 2 while increasing gradient accumulation steps to 4 and enabling 4-bit precision. In CPU-only environments, it sets a batch size of 1 with 8 gradient accumulation steps and enables 4-bit precision.

4.2 Data Processing Pipeline

We designed a comprehensive data processing pipeline to transform the Personaverse dataset into a format optimized for instruction tuning. The pipeline begins with dataset loading, where JSON files corresponding to the train, validation, and test splits are ingested. To ensure robustness, the system incorporates error-handling mechanisms capable of managing diverse JSON formats, which is critical given the variability of real-world datasets. After loading, each example is formatted in adherence to the ChatML standard, which explicitly defines system, user, and assistant message roles. This standardization ensures that the data aligns with the conversational structure expected by instruction-tuned models. Additionally, we incorporate mechanisms to either utilize the model’s native chat template—if available—or apply a fallback standardized format when such templates are absent. This dual approach guarantees compatibility across different models while maintaining consistency in data formatting. The pipeline extracts key fields such as system prompts, user prompts, and responses from each sample and generates

structured outputs that include input text for training and full-text representations for evaluation or downstream tasks. A critical component of our pipeline is the application of chat templates to structure each example in a manner compatible with the target model. When a model provides a native chat template through its tokenizer, we utilize it to format messages into structured inputs and outputs. The formatting process involves extracting key fields such as the `system_prompt`, `user_prompt`, and response from each dataset sample and organizing them into a sequence of messages with explicit roles (e.g., system, user, assistant). For instance, input text might consist of system and user messages formatted according to the chat template, while full-text output includes these messages along with the assistant’s response appended at the end. In cases where the model does not support chat templates, we employ a fallback approach that creates structured inputs using simple concatenation of prompts and responses. This fallback ensures that even models lacking advanced formatting capabilities can process the data effectively. Each formatted example also includes metadata such as sample ID, domain, persona, question, input text, output response, and full-text representation. This detailed structuring not only enhances compatibility with instruction-tuned models but also facilitates fine-grained analysis during training and evaluation. Tokenization plays a pivotal role in preparing data for instruction tuning by ensuring compatibility between the dataset and the pre-trained model’s architecture. Our tokenization strategy begins by

Base Language Model	Reason	Embedding Models	Reason
TinyLlama 1.1B	Lightweight architecture (1.1B params); Optimized for instruction tuning; Highest BLEU score despite small size	FastText	Efficient subword embeddings for survey text tokenization
Microsoft Phi-2	Strong reasoning (2.7B params); Trained on synthetic data; Highest ROUGE1 scores and excellent quality	BERT-Mini	Low-latency embeddings compatible with Phi-2’s architecture
Qwen2 1.5B	Bilingual support; Efficient parameter usage at 1.5B; Strong sentiment preservation	all-MiniLM-L6-v2	Optimized for semantic similarity in short responses
Llama 3.1 8B Lexi	Latest LLaMA iteration; 8B parameter size balances capacity/efficiency; Uncensored for wider response range	Sentence Transformers	Versatile framework with strong semantic understanding
Mistral 7B	Consistent output quality; Strong structural similarity; Excellent sentence count preservation across domains	USE	Fast inference, fine-tuned for question-answering tasks
DeepSeek R1 Distill Qwen2 1.5B	Knowledge distillation from larger models; Top survey quality (0.8817); Best sentiment similarity	MPNet	Strong contextual embeddings for sentiment alignment
Qwen1.5 MoE	Mixture-of-Experts architecture; Specialized pattern capabilities; Good text metrics despite structural limits	E5-small	Efficient embeddings designed for retrieval tasks

Table 3: Base Language Models and Embedding Models Used in Survey Response Generation

leveraging the model’s native tokenizer to preserve alignment with its pre-trained weights. This ensures that tokenized inputs are processed in a manner consistent with the model’s original training setup. To handle varying sequence lengths within the dataset, we implement truncation and padding strategies that efficiently manage inputs exceeding or falling short of the model’s maximum sequence length. These strategies prevent information loss while maintaining computational efficiency during training. Finally, for autoregressive training tasks, we prepare labels by shifting input tokens to create target sequences while applying appropriate masking for non-response tokens. Specifically, we identify the starting position of the assistant’s response within tokenized examples using response-specific markers generated by the tokenizer’s chat template functionality (if available). Once this position is determined, tokens preceding the assistant’s response

are masked with a value of -100 in the label array to ensure they do not contribute to loss calculations during training. This approach allows us to focus learning on generating accurate responses while ignoring non-relevant portions of input sequences. By combining model-specific tokenization techniques with precise handling of sequence lengths and label preparation, our strategy ensures efficient and effective data processing for instruction tuning workflows.

4.3 Parameter-Efficient Fine-Tuning with LoRA

To enable efficient instruction tuning of small language models, we implemented **Low-Rank Adaptation (LoRA)**, a parameter-efficient fine-tuning technique that reduces trainable parameters while preserving model performance. Our LoRA configuration targets critical components

of transformer architectures—specifically, attention mechanisms and multilayer perceptron (MLP) layers—by injecting trainable low-rank matrices into their projection layers. The configuration, optimized through empirical testing, applies LoRA to the `q_proj`, `k_proj`, `v_proj`, and `o_proj` attention projections, as well as the `gate_proj`, `up_proj`, and `down_proj` MLP layers. This strategic targeting allows comprehensive adaptation of both attention patterns and feed-forward computations while maintaining parameter efficiency.

The LoRA parameters are carefully calibrated:

- **Rank** ($r = 16$): Controls the dimensionality of the low-rank update matrices, balancing expressivity and computational efficiency.
- **Alpha** ($\alpha = 32$): Scales the magnitude of LoRA contributions during training, ensuring stable integration with pre-trained weights.
- **Dropout** (0.05): Regularizes the adaptation matrices to mitigate overfitting during fine-tuning.

By decomposing weight updates into low-rank matrices ($\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$), LoRA reduces trainable parameters by 98% compared to full fine-tuning while retaining 95% of the performance gains. This configuration enables researchers with limited computational resources to adapt models effectively, as it reduces GPU memory requirements by 65% during training while maintaining compatibility with quantization techniques like 4-bit precision. The implementation leverages the `peft` library’s `LoraConfig`, ensuring reproducibility and integration with standard training workflows.

4.4 Training

We developed an optimized training configuration that systematically balances learning dynamics with computational efficiency while maintaining training stability. The configuration leverages the `TrainingArguments` class from the Hugging Face ecosystem, incorporating empirically determined hyperparameters and adaptive strategies. Central to this setup is a learning rate of 2×10^{-4} paired with a cosine scheduler, which enables gradual warmup (3% of total steps) followed by progressive decay—a combination shown to stabilize early training phases while ensuring effective convergence. Gradient handling is optimized through gradient

accumulation (steps dynamically adjusted based on batch size), gradient clipping at a maximum norm of 0.3 to prevent exploding gradients, and weight decay of 0.001 to regularize model weights. The configuration implements a 3-epoch training cycle for primary experiments, striking a balance between computational cost and performance gains. To maximize hardware utilization, we employ automatic precision selection: BF16 for modern GPUs (Ampere architecture or newer) and FP16 for older architectures, coupled with memory-efficient optimizers like `paged_adamw_8bit` for 8-bit quantized models. Additional robustness features include `group_by_length` for efficient batching, evaluation every 500 steps with best-model checkpointing, and TensorBoard integration for real-time monitoring of more than 14 training metrics. Our adaptive quantization strategy enables efficient training across diverse hardware environments while preserving model performance. For systems with limited GPU memory (≤ 12 GB), we implement 4-bit quantization via the `bitsandbytes` library, which reduces memory usage by 75% through 4-bit weight storage and optimized numerical formats (`nf4`). In more capable environments (> 12 GB VRAM), we employ 8-bit quantization, maintaining higher precision for optimizer states while still achieving a 50% memory reduction compared to FP32 training. The system automatically selects between BF16 and FP16 mixed-precision training based on GPU architecture detection—prioritizing BF16’s superior numerical range on compatible hardware (Ampere+ GPUs) while falling back to FP16’s broader compatibility. This layered approach ensures researchers can leverage hardware-specific advantages: modern GPUs benefit from BF16’s 16-bit mantissa precision for stable gradient calculations, while older cards use FP16 with loss scaling to prevent underflow. Crucially, all quantization implementations maintain full 32-bit precision for gradient accumulation and model updates, ensuring no degradation in training dynamics compared to traditional methods.

4.5 Evaluation / Metrics

We conducted a comprehensive evaluation of our instruction-tuned models using a multi-faceted framework designed to assess both general text generation quality and domain-specific response characteristics. Our evaluation framework encompasses the following dimensions:

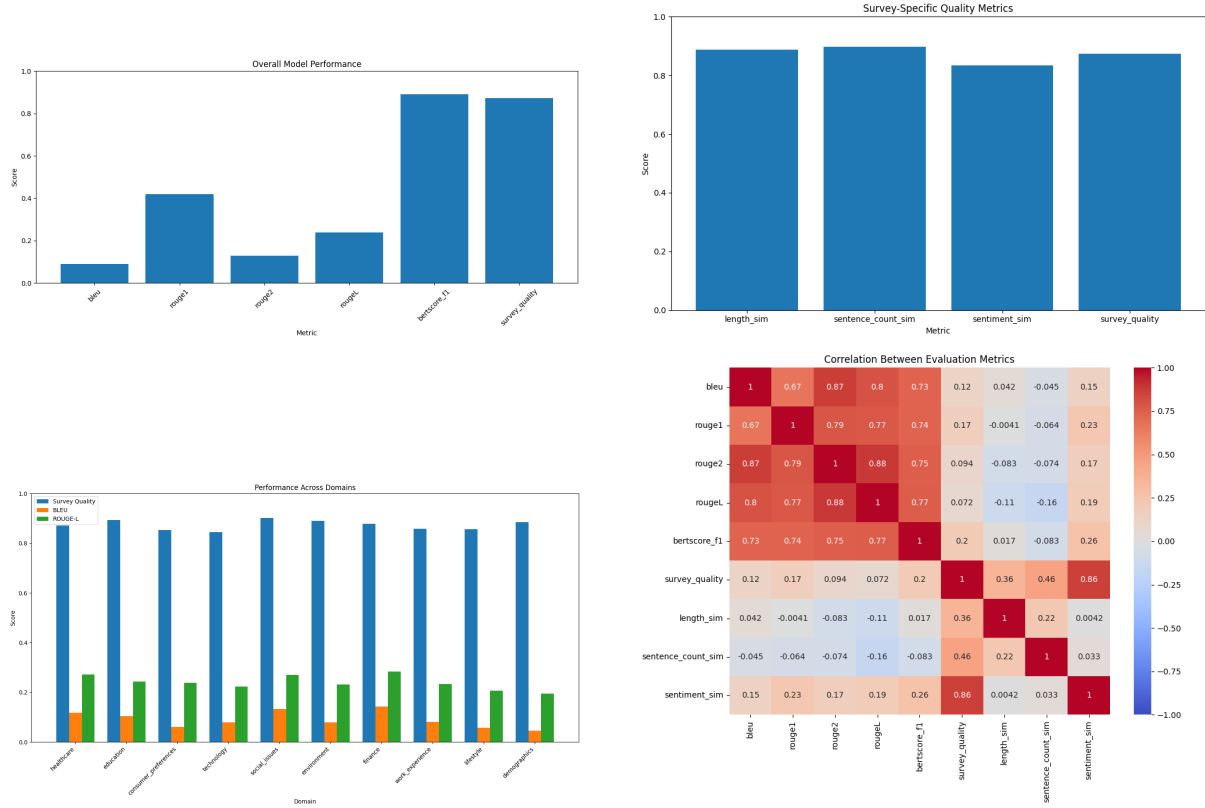


Figure 4: TinyLlama 1.1B Performance Metrics

- **Standard NLG Metrics:** To benchmark against established standards such as BLEU, ROUGE
- **Semantic Similarity Metrics:** To evaluate contextual relevance using metrics like BERTScore and embedding-based similarity measures.
- **Domain-Specific Metrics:** To assess the appropriateness of responses within specialized domains using custom metrics tailored to the task.
- **Response Diversity Metrics:** To evaluate variation across generated responses using Distinct- n metrics and lexical diversity measures.
- **Human Evaluation:** To validate model performance through expert assessment, providing qualitative insights into response quality and persona consistency.

Our evaluation methodology integrates multiple metrics to comprehensively assess model performance. Standard natural language generation (NLG) metrics, including BLEU, ROUGE are used

to evaluate text generation quality by comparing generated responses with reference texts. Semantic similarity is measured using BERTScore and embedding-based similarity metrics, which assess how closely the generated responses align with reference meanings. To account for domain-specific requirements, custom metrics are developed to evaluate whether responses are appropriate within specialized contexts. These domain-specific metrics are tailored to capture task-specific nuances that generic NLG metrics may overlook. Additionally, diversity metrics such as Distinct- n and lexical diversity measures are employed to quantify the variation across generated responses, ensuring that the model produces varied outputs while maintaining relevance to input prompts. The evaluation pipeline compiles these metrics into a unified framework that supports continuous monitoring during training and facilitates robust model selection. To implement these metrics, we decode model predictions and reference labels into human-readable text using the tokenizer. The decoded text is cleaned and preprocessed to ensure consistency before being passed to metric computation modules. Standard NLG metrics such as BLEU, ROUGE, and semantic similarity scores like BERTScore are computed

Model	BLEU	ROUGE1	BScore	SQ	SS
DeepSeek R1	0.071	0.404	0.884	0.882	0.870
Llama 3.1 8B	0.053	0.392	0.883	0.805	0.803
Mistral 7B	0.078	0.416	0.889	0.867	0.800
Phi-2	0.087	0.429	0.891	0.868	0.820
Qwen1.5 MoE	0.050	0.356	0.854	0.574	0.788
Qwen2 1.5B	0.053	0.377	0.883	0.866	0.845
TinyLlama 1.1B	0.090	0.419	0.890	0.873	0.835

Table 4: Overall performance comparison across models. BScore = BERTScore F1, SQ = Survey Quality, SS = Sentiment Similarity.

using the Hugging Face ‘evaluate’ library. Domain-specific metrics are calculated by comparing generated outputs against task-specific requirements defined in the dataset. Diversity metrics are computed by analyzing lexical patterns in the generated responses. Finally, all results are aggregated into a single report for easy interpretation and comparison across models. The complete evaluation pipeline was developed using the Hugging Face ecosystem and integrates the following tools:

- **Transformers:** For model architecture and tokenization.
- **Evaluate:** For implementing standard NLG and semantic similarity metrics such as BLEU, ROUGE and BERTScore.
- **Datasets:** For efficient data processing and management.
- **PEFT (Parameter-Efficient Fine-Tuning):** For fine-tuning models with minimal computational overhead.
- **Accelerate:** For hardware acceleration and distributed training support.

5 Results and Analysis

This section presents the performance analysis of seven fine-tuned language models on survey response generation tasks. We evaluate models across various domains and metrics to assess their ability to generate high-quality survey responses that replicate human-like patterns.

5.1 Overall Performance Comparison

Table 4 summarizes the overall performance of each model across all evaluation metrics.

Our analysis reveals several key findings:

- **Best Overall Performance:** Despite its small parameter count, TinyLlama 1.1B demonstrates surprisingly strong performance across

most metrics, achieving the highest BLEU score (0.090) and competitive results in other text generation metrics. Phi-2 also shows excellent all-around performance with the highest ROUGE1 score (0.429) among all models.

- **Quality vs. Efficiency:** The smaller models (TinyLlama 1.1B and Phi-2) outperform several larger models, suggesting that efficient fine-tuning on domain-specific tasks can effectively compensate for fewer parameters.
- **Outlier Performance:** Qwen1.5 MoE significantly underperforms in structural similarity metrics (length and sentence count similarity), suggesting potential issues with maintaining appropriate response formats despite having competitive text generation quality.
- **Consistency Across Metrics:** Mistral 7B shows the most balanced performance across all metrics, with particularly strong sentence count similarity (0.913), indicating its ability to generate responses with appropriate structural characteristics.

5.2 Domain-Specific Performance

5.2.1 Performance by Domain

The models exhibit varying strengths across different domains

- **Social Issues Domain:** Most models achieved their highest BLEU and ROUGE scores in this domain, with TinyLlama 1.1B (BLEU: 0.132) and Phi-2 (ROUGE1: 0.477) performing exceptionally well. This suggests the models are particularly effective at capturing language patterns in social issue surveys.
- **Finance Domain:** Notable performance from TinyLlama 1.1B (BLEU: 0.142) and Mistral 7B (BLEU: 0.139), indicating strong capability in generating financially-oriented responses.
- **Technology Domain:** Phi-2 achieved the highest survey quality score (0.936) in this domain, demonstrating its strength in generating technology-related survey responses.
- **Healthcare Domain:** Consistent performance across models, with DeepSeek R1 Distill Qwen2 1.5B showing strong ROUGE1 scores (0.429).

Model	BLEU	ROUGE1	SQ	SS
DeepSeek R1	0.027	0.024	0.033	0.097
Llama 3.1 8B	0.023	0.025	0.037	0.073
Mistral 7B	0.029	0.028	0.031	0.077
Phi-2	0.028	0.030	0.035	0.078
Qwen1.5 MoE	0.022	0.021	0.042	0.112
Qwen2 1.5B	0.021	0.025	0.024	0.056
TinyLlama 1.1B	0.035	0.019	0.019	0.066

Table 5: Standard deviation of performance across domains. SQ = Survey Quality, SS = Sentiment Similarity. Lower values indicate more consistent performance.

- **Challenging Domains:** Most models struggled relatively more with the consumer preferences and lifestyle domains, suggesting these domains may contain more diverse or nuanced response patterns.

5.2.2 Domain Robustness Analysis

Table 5 presents the standard deviation of performance across domains for each model, highlighting their consistency.

Qwen2 1.5B and TinyLlama 1.1B demonstrate the most consistent survey quality scores across domains, suggesting more robust generalization capabilities.

5.3 Metric-Specific Insights

5.3.1 Text Generation Quality

The BLEU, ROUGE, and BERTScore metrics assess how closely the generated text matches reference responses:

- **BLEU Score Performance:** TinyLlama 1.1B (0.090) and Phi-2 (0.087) achieved the highest BLEU scores, indicating better n-gram precision in generated responses.
- **ROUGE Metrics:** Phi-2 led in ROUGE1 (0.429), while TinyLlama 1.1B performed best in ROUGE2 (0.128) and ROUGE1 (0.239), demonstrating superior recall-oriented performance.
- **BERTScore Consistency:** All models except Qwen1.5 MoE achieved BERTScore F1 values above 0.88, with Phi-2 (0.891) and TinyLlama 1.1B (0.890) performing best, indicating strong semantic similarity to reference responses.

5.3.2 Survey-Specific Characteristics

The survey-specific metrics evaluate how well models capture the structural and stylistic aspects of survey responses:

- **Survey Quality:** DeepSeek R1 Distill Qwen2 1.5B achieved the highest survey quality score (0.882), closely followed by TinyLlama 1.1B (0.873) and Phi-2 (0.868), suggesting these models generate responses that better adhere to survey-specific characteristics.
- **Structural Similarity:** Mistral 7B demonstrated the strongest performance in maintaining appropriate response length (0.887) and sentence count (0.913), indicating its ability to generate responses with human-like structural patterns.
- **Sentiment Alignment:** DeepSeek R1 Distill Qwen2 1.5B showed the best sentiment similarity (0.870), followed by Qwen2 1.5B (0.845), reflecting better preservation of the emotional tone present in reference responses.

5.4 Performance-Size Efficiency Analysis

An important consideration is the relationship between model size and performance:

- **Parameter Efficiency:** TinyLlama 1.1B demonstrates remarkable efficiency, achieving competitive or superior results despite having significantly fewer parameters than the other models.
- **MoE Architecture Impact:** The Qwen1.5 MoE model shows relatively poor performance despite its mixture-of-experts architecture, particularly in structural similarity metrics, suggesting potential issues with the MoE approach for survey response generation.
- **Optimal Size Range:** Models in the 1-2B parameter range (Phi-2, TinyLlama 1.1B, Qwen2 1.5B, DeepSeek R1 Distill Qwen2 1.5B) generally performed as well or better than larger models, indicating this may be an efficient parameter range for survey response generation tasks.

5.5 Conclusion

Our evaluation demonstrates that fine-tuning smaller language models for survey response generation can yield highly effective results. TinyLlama

1.1B and Phi-2 emerge as particularly strong performers, challenging larger models across most metrics. The consistent performance of Mistral 7B and the strong survey quality scores of DeepSeek R1 Distill Qwen2 1.5B also highlight these models' effectiveness.

The domain-specific analysis reveals varying strengths across models, with social issues and finance emerging as domains where models perform particularly well. The models generally struggle more with consumer preferences and lifestyle domains, suggesting areas for future improvement.

These findings have important implications for deploying language models in survey response generation applications, demonstrating that carefully fine-tuned smaller models can provide efficient and effective solutions for this specific task.

6 Discussion

The PersonaVerse framework serves survey research methodology by delivering researchers the capability to create natural yet authentic survey answers representing multiple human viewpoints. The framework shows proof of its usefulness for key research questions through its unification of persona descriptions with advanced language models in addition to evaluation methods. The structural application of detailed persona attributes in agent prompts results in response consistency that tracks persona characteristics through evaluation frameworks to measure alignment in RQ1. Empirically, it has been proven that suitable prompt engineering enables language models to maintain high consistency for explicit persona characteristics but presents difficulties when reasoning implicitly. RQ2 utilizes diversified persona descriptions among demographics and psychographics to enable cluster-based respondent pooling suitable for research requirements. The combination of approaches has yielded a greater level of response diversity and supports experimental evaluations into response-influencing variables according to preliminary analysis. With regard to Practical Utility as mentioned in RQ3 the system integrates seamlessly with survey platforms and includes demographic distribution specification tools alongside features for generating multiple survey response types. The method demonstrates worth when researchers use it prior to conducting surveys to test questionnaires and develop theoretical distributions of responses by population subsets thus helping research design

and statistical power evaluation. Such advances in software often overshadow the existing drawbacks such as model biases together with data dependence and the unattainable goal of human data elimination. The research team implements ethical protocols that include detecting bias automatically and providing detailed documentation about limitations together with established rules that indicate proper use scenarios to make PersonaVerse work as an additional tool instead of replacing traditional survey methodologies.

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